Toxicology and Pharmacological Interactions of Classic Psychedelics

Kelan Thomas

Contents

1 Introduction
2 LSD
  2.1 LSD Toxicology
  2.2 LSD Drug–Drug Interactions
3 Psilocybin
  3.1 Psilocybin Toxicology
  3.2 Psilocybin Drug–Drug Interactions
4 DMT
  4.1 DMT Toxicology
  4.2 DMT Drug–Drug Interactions
5 5-MeO-DMT
  5.1 5-MeO-DMT Toxicology
  5.2 5-MeO-DMT Drug–Drug Interactions
6 Mescaline
  6.1 Mescaline Toxicology
  6.2 Mescaline Drug–Drug Interactions
7 2C-B
  7.1 2C-B Toxicology
  7.2 2C-B Drug–Drug Interactions
8 Bromo-DragonFLY
  8.1 Bromo-DragonFLY Toxicology
  8.2 Bromo-DragonFLY Drug–Drug Interactions
9 25X-NBOMe
  9.1 25X-NBOMe Toxicology
  9.2 25X-NBOMe Drug–Drug Interactions
10 Psychedelics and Serotonin Toxicity
11 Psychedelic Long-Term Microdosing and Valvular Heart Disease Risk
12 Discussion and Conclusion
References

K. Thomas (✉)
Clinical Sciences, Touro University California College of Pharmacy, Vallejo, CA, USA
e-mail: kthomas10@touro.edu

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2024
Curr Topics Behav Neurosci
https://doi.org/10.1007/7854_2024_508

Abstract As psychedelics are being investigated for more medical indications, it has become important to characterize the adverse effects and pharmacological interactions with these medications. This chapter will summarize what is known about the toxicology and drug–drug interactions for classic psychedelics, such as LSD, psilocybin, DMT, 5-MeO-DMT, mescaline, 2C-B, Bromo-DragonFLY, and 25X-NBOMe.

Keywords Adverse effects · Drug–drug interactions · Metabolism · Psychedelic · Toxicology

1 Introduction

Classic psychedelics, such as LSD, psilocybin, DMT, 5-MeO-DMT, mescaline, 2C-B, Bromo-DragonFLY, and 25X-NBOMe compounds, are primarily active at the 5-HT_{2A} receptor to induce psychedelic subjective effects. However, each individual compound may have slightly different functional selectivity and second messenger signaling at the 5-HT_{2A} receptor, along with activity at other targets relevant to consider for their toxicological and pharmacological interaction potential (Nichols 2016). Pharmacokinetic clinical trials will also be necessary to determine the clinical significance of Phase I and II metabolic enzymes related to pharmacogenetics and drug–drug interactions of these psychedelics.

This chapter will summarize what is known about the toxicology and pharmacological interactions for each psychedelic compound from clinical trials, epidemiologic studies, and case reports.

2 LSD

LSD, lysergic acid diethylamide, is physiologically safe at usual doses of 50–200 μg, and there have been about 1,000 clinical case reports of approximately 40,000 individuals in the medical literature published during the 1950s and 1960s (Nichols and Grob 2018). Oral LSD has an estimated bioavailability of 71% and plasma concentrations peak around 1.5 h, while subjective effects last about 8–12 h and the elimination half-life is approximately 3 h (Libânio Osório Marta 2019). LSD metabolism has not been well-characterized, but studies suggest that CYP1A2, CYP2C9, CYP2C19, CYP3A4, and CYP2E1 may contribute to the formation of LSD's primary metabolite 2-Oxo-3-hydroxy-LSD (O-H-LSD), while CYP2D6, CYP3A4, and CYP2E1 may also contribute to formation of a more minor metabolite N-desmethyl-LSD (nor-LSD) (Wagmann et al. 2019; Libânio Osório Marta 2019). In a study of healthy volunteers, participants with no functional CYP2D6 alleles (poor metabolizers) had roughly 75% higher exposure (AUC) compared to normal

CYP2D6 metabolizers, which suggests that CYP2D6 inhibitors may also theoretically increase LSD exposure (Vizeli et al. 2021). LSD glucuronides detected in human samples suggest that UGT enzymes are the most important Phase II metabolism reaction in humans (Libânio Osório Marta 2019).

2.1 LSD Toxicology

A dose of LSD 100 mg freebase is the extrapolated usual lethal dose based on animal studies and human case reports, which is 1,000 times greater than an average 100 μg dose consumed (Gable 2004). The very few case reports that have implicated LSD in fatalities have been critically evaluated and two were likely related to a massive overdose (possibly >300 mg by extrapolation), the other 3 cases were more likely related to adulterants or physical restraint by police rather than LSD itself (Nichols and Grob 2018). In a notorious clinical report of extremely high LSD dosages, 8 people accidentally insufflated two "lines" of LSD tartrate powder and all presented to the emergency room about 15 min later with serotonin toxicity symptoms of sinus tachycardia, mydriasis, emesis, and diaphoresis (Klock et al. 1975). Five were comatose upon initial medical evaluation, 4 had fever, 4 had mild generalized bleeding, 3 required intubation and assisted ventilation, 3 aspirated vomitus, 3 had hypertension, 2 had diarrhea, and 1 had hyperreflexia (Klock et al. 1975). Labs upon admission revealed that all samples had coagulopathy, while LSD blood concentrations ranged from 2.1 to 26 ng/mL, and after 12 h all of them had completely recovered with supportive care, respiratory assistance, IV fluids, and hypothermic blankets (Klock et al. 1975). A retrospective analysis of 3,554 single substance LSD exposures, reported to poison centers between 2000 and 2016, found that serious clinical effects were infrequent and included the following: 3.8% fever/hyperthermia, 2.4% single seizure, 1.4% coma, 1.4% increased creatinine, 1.2% multi-discrete seizures, 1.1% rhabdomyolysis, 0.9% respiratory depression, 0.5% cardiac conduction disturbance, and 0.4% status epilepticus (Leonard et al. 2018).

2.2 LSD Drug–Drug Interactions

Drug inhibitors of CYP3A4 and CYP1A2 decreased individual LSD metabolite formation by approximately 85% and 60%, respectively, when compared to in vitro enzyme incubations without an inhibitor (Wagmann et al. 2019). Another in vitro study also determined that CYP3A4, which has many strong inhibitors, was quantitatively the most important enzyme for LSD metabolism to both O-H-LSD and nor-LSD (Luethi et al. 2019). Inhibitors of CYP3A4, CYP2E1, CYP2D6, CYP1A2, and CYP2C9 all decreased LSD metabolite formation, while CYP1A2 and CYP3A4 inducers like omeprazole and rifampicin increased LSD metabolite formation by approximately 30%, suggesting that LSD may be susceptible to drug–drug

interactions and genetic polymorphism effects (Luethi et al. 2019). Clinical trials with drug inhibitors/inducers and genetic polymorphisms are necessary to elucidate if these in vitro metabolic changes result in clinically significant alterations in drug exposure.

Ketanserin is an antihypertensive, that is also an antagonist at the 5-HT_{2A} receptor, which drastically attenuated LSD subjective effects by 60–70% as measured by visual analog scale area under the effect curves (VAS AUEC) (Becker et al. 2022b). Ketanserin 40 mg pretreatment 1 h prior to LSD 100 μg and 200 μg effectively prevented the psychedelic effects, while ketanserin 40 mg administered 1 h after LSD 100 μg also reduced the duration of psychedelic subjective effects from 8.5 h with placebo to only 3.5 h with ketanserin (Holze et al. 2021; Olbrich et al. 2021; Becker et al. 2022b).

Ergotamine, an antimigraine 5-HT_{2A} agonist medication, pretreatment with 2–4 mg for 3–5 days attenuated LSD central sympathetic activity as measured by lower pulse rate and less pupillary dilation, but also prolonged and intensified psychedelic subjective effects like excitation and euphoria (Mattusek and Halbach 1964).

Chlorpromazine, an antipsychotic with relatively strong binding affinity for 5-HT_{2A} (Ki = 2 nM), pretreatment dosed at 25–100 mg dosed 30 min prior to LSD 15–60 μg also reduced the intensity of psychedelic subjective effects (Isbell 1957; Murphree 1962; Maheux et al. 2005).

Isocarboxazid, a monoamine oxidase inhibitor (MAOI) antidepressant, pretreatment with 30 mg daily for 2 and 5 weeks markedly attenuated LSD 40 μg response, including psychological effects, autonomic effects, and neurological responses, where some participants no longer noticed any discernible effect from LSD (Resnick et al. 1964). Two cases of people who took LSD before and after chronic monoamine oxidase inhibitor (MAOI) antidepressant exposure also had attenuated LSD subjective effects, which may theoretically be related to MAOI-induced 5-HT_{2A} receptor desensitization, while 5 people who took LSD before and after chronic tricyclic antidepressant (TCA) exposure actually noticed an increased overall response to LSD (Bonson and Murphy 1995; Van Oekelen et al. 2003). A case series of people who took LSD before and after chronic selective serotonin reuptake inhibitor (SSRI) antidepressant exposure demonstrated that SSRIs attenuated LSD subjective effects, which may also theoretically be related to 5-HT_{2A} receptor desensitization (Bonson 1996; Yamauchi et al. 2006). These opposite effects between chronic TCA and SSRI antidepressant exposure suggest there may be more complex pharmacodynamic and pharmacokinetic drug–drug interactions involved that should be evaluated in controlled clinical trials with TCAs (Bonson and Murphy 1995; Bonson 1996).

Two people who took LSD before and after chronic lithium exposure had increased LSD subjective effects in this case series, while another study found that 49% of the 55 LSD plus lithium online trip reports reported seizures, including 39% of these reports seeking emergency medical treatment (Bonson and Murphy 1995; Nayak et al. 2021).

3 Psilocybin

Psilocybin, O-phosphoryl-4-hydroxy-*N,N*-dimethyltryptamine (active metabolite psilocin), is physiologically safe at usual doses of 10–30 mg, and there have been about 100 reported studies in humans (Nichols 2020). Oral psilocybin is immediately dephosphorylated to psilocin, which has an estimated bioavailability of 53% and plasma concentration peak around 1.5 h, while subjective effects last about 4–6 h and the elimination half-life is approximately 3 h (Hasler et al. 1997; Dinis-Oliveira 2017; Brown et al. 2017). Psilocybin metabolism has not been well characterized, especially as it relates to Phase I metabolism, but in vitro studies suggest that glucuronidation by UGT1A9 and UGT1A10 enzymes is likely the most important Phase II metabolism reaction in humans (Manevski et al. 2010; Dinis-Oliveira 2017).

3.1 Psilocybin Toxicology

A dose of psilocybin 6 g is the extrapolated usual lethal dose based on animal studies and one human case report, which is 200 times greater than a typical 30 mg dose consumed in clinical trials (Gable 2004; Nichols 2020). The few case reports that have implicated psilocybin mushrooms in fatalities have been reviewed, with two deaths considered due to overdosing with magic mushrooms and another case of a 6-year-old child who developed status epilepticus with hyperthermia potentially related to *Psilocybe baeocystis* ingestion (Amsterdam et al. 2011). There is also another fatal case report of a 24-year-old heart transplant recipient who presented with cardiac arrest 2–3 h after ingesting an unknown quantity of magic mushrooms (Lim et al. 2012). A retrospective analysis of 5,883 single substance psilocybin mushrooms exposures, reported to poison centers between 2000 and 2016, found that serious clinical effects were infrequent and included the following: 2.0% fever/hyperthermia, 0.6% single seizure, 0.4% coma, 0.4% increased creatinine, 0.3% rhabdomyolysis, 0.2% multi-discrete seizures, 0.2% respiratory depression, and 0.2% cardiac conduction disturbance (Leonard et al. 2018).

3.2 Psilocybin Drug–Drug Interactions

Drug inhibitors of UGT1A9 and UGT1A10 enzymes, such as diclofenac or probenecid, may theoretically increase exposure to psilocin, but there are no studies that have evaluated this hypothetical drug–drug interaction (Sarparast et al. 2022).

There have been several clinical trials evaluating pharmacodynamic drug–drug interactions to determine how psychiatric medications may alter psilocybin effects. Similar to LSD, medications that compete with psilocybin at the 5-HT_{2A} receptor

would be expected to attenuate psilocybin subjective effects and this has been demonstrated in drug–drug interaction clinical trials.

Ketanserin pretreatment 75 min prior to psilocybin 0.25 mg/kg reduced psilocybin subjective effects in a dose-dependent manner; ketanserin 20 mg reduced visionary restructuralization by 20%, dread of ego-dissolution by 15%, and oceanic boundlessness by 16%, while ketanserin 40 mg reduced visionary restructuralization by 32%, dread of ego-dissolution by 21%, and oceanic boundlessness by 29% as measured by Altered States of Consciousness Rating Scale (APZ-OAV) (Vollenweider et al. 1998). Ketanserin pretreatment with 50 mg prior to psilocybin 0.215 mg/kg more dramatically reduced (>50%) visionary restructuralization, anxious ego-dissolution, and oceanic boundlessness, but did not significantly change auditory alterations on the 5-Dimensional Altered States of Consciousness Rating Scale (5D-ASC) (Carter et al. 2005, 2007).

Risperidone, the atypical antipsychotic with potent 5-HT_{2A} antagonism, pretreatment dosed at 0.5 mg reduced psilocybin subjective effects like visionary restructuralization by 19% and dread of ego-dissolution by 22% as measured with the APZ-OAV (Vollenweider et al. 1998). Risperidone 1 mg pretreatment also reduced psilocybin subjective effects like visionary restructuralization by 27%, dread of ego-dissolution by 25%, and oceanic boundlessness by 31% measured by APZ-OAV (Vollenweider et al. 1998).

While most typical antipsychotics have insignificant affinity for 5-HT_{2A} receptor, some typical antipsychotics do have relatively strong binding affinity at the 5-HT_{2A} receptor, such as haloperidol (Ki = 60 nM) and chlorpromazine (Ki = 2 nM), which could compete with psilocybin 5-HT_{2A} binding (Maheux et al. 2005). Chlorpromazine 50 mg pretreatment reduced psilocybin subjective effects, like changes in after-image effect by 95% and major unusual visual experience by 53% (Keeler 1967). In the same study that investigated risperidone, haloperidol 0.021 mg/kg intravenous pretreatment reduced psilocybin subjective effect oceanic boundlessness by 14% and increased dread of ego-dissolution by 27% on APZ-OAV, suggesting that haloperidol was even more dysphoric than risperidone pretreatment with disordered thoughts and paranoia (Vollenweider et al. 1998).

Buspirone, an anxiolytic 5-HT_{1A} partial agonist, also has clinically relevant binding affinity at the 5-HT_{2A} receptor (Ki =138 nM) and buspirone 20 mg pretreatment 100 min prior to psilocybin reduced subjective effect of visionary restructuralization by 44% as measured on the 5D-ASC (Bonhaus et al. 1997; Pokorny et al. 2016). Ergotamine pretreatment 100 min prior to psilocybin did not significantly alter subjective effects on 5D-ASC (Pokorny et al. 2016).

Escitalopram is an SSRI that has negligible affinity for 5-HT_{2A}, but 2 weeks of serotonin reuptake inhibitor pretreatment with 10 mg for 7 days and 20 mg for 7 days may theoretically desensitize the receptor, which reduced psilocybin subjective effects like anxious ego-dissolution by 40% as measured on the 5D-ASC and ineffability by 19% as measured by Mystical Experience Questionnaire (MEQ30), along with attenuating psilocybin-related increases in blood pressure by 4–7 mmHg (Becker et al. 2022a).

4 DMT

DMT, *N,N*-dimethyltryptamine, is physiologically safe at usual doses, 40–50 mg for freebase DMT inhalation and 25–60 mg for DMT content consumed orally in an ayahuasca brew. Injected doses of DMT are usually between 0.1 and 0.4 mg/kg and doses around 0.2 mg/kg for both inhaled and injected DMT are associated with threshold psychedelic subjective effects (Brito-da-Costa et al. 2020). Injected and inhaled DMT peaks around 5 min and wears off in less than an hour, while DMT orally ingested in ayahuasca brew typically peaks after 1.5–2 h and subjective effects last about 4–6 h with an estimated oral elimination half-life of 1–4 h (Brito-da-Costa et al. 2020). Oral DMT contained in *Psychotria viridis* and *Diplopterys cabrerana* leaves undergoes extensive first-pass metabolism to 3-indole-acetic acid (3-IAA) and therefore has very poor bioavailability, but when combined with MAOI harmala alkaloids from the stem and bark of *Banisteriopsis caapi*, which peak after 1.5–3 h and have elimination half-lives between 2 and 9 h, then enough DMT is absorbed that can eventually cross the blood brain barrier for psychoactive effects (Brito-da-Costa et al. 2020). This is because the beta-carboline alkaloids harmine and harmaline (and to some extent tetrahydroharmine) are reversible inhibitors of monoamine oxidase A enzymes (RIMAs), which may pose a risk of serotonin toxicity when combined with medications that also increase synaptic serotonin concentration (Malcolm and Thomas 2021). While MAO-A is the primary metabolic pathway for DMT to 3-IAA, there is also Phase I metabolism by CYP enzymes to DMT-*N*-oxide (Brito-da-Costa et al. 2020). CYP1A2 and CYP2D6 are thought to be the major enzymes for metabolism of harmala alkaloids, but harmine is also metabolized by CYP2C9 and CYP2C19 (Brito-da-Costa et al. 2020).

4.1 DMT Toxicology

A dose of oral DMT 2 g is the extrapolated usual lethal dose based on animal studies, which is 50 times greater than an average 40 mg usual effective dose consumed (Gable 2004). There have been a few fatal case reports associated with an ayahuasca brew, a decoction of plants containing DMT and β-carboline harmala alkaloids such as harmine and tetrahydroharmine (THH), but these reports also had co-ingestions of additional substances thereby making it difficult to determine any causal association (dos Santos 2013). In one fatal case report a 71-year-old female was given an enema of the ayahuasca preparation that also contained nicotine, and her nicotine concentration was at least 20 times higher than average postmortem concentration and was considered the cause of death (dos Santos 2013). In another fatal case report a 25-year-old male was found deceased after consuming unknown herbal extracts and postmortem toxicology identified blood concentrations of harmine, THH, harmaline, DMT, 5-MeO-DMT, and diphenhydramine, so it would also be inaccurate to attribute this fatal case report specifically to only the traditional ayahuasca brew (dos Santos 2013).

4.2 DMT Drug–Drug Interactions

DMT is metabolized by MAO-A, CYP1A2, and CYP2C19, so theoretically drugs that inhibit these enzymes could increase DMT concentrations, but this has not been evaluated in clinical trials (Riba et al. 2012; Wagmann et al. 2016). Pretreatment with pindolol 30 mg, a beta-blocker antihypertensive with 5-HT_{1A} affinity, prior to IV DMT 0.1 mg/kg intensified psychedelic effects by two to three times as measured on the Hallucinogen Rating Scale, which was purportedly caused by pindolol 5-HT_{1A} receptor competition and subsequent unopposed DMT 5-HT_{2A} activity (Strassman 1995). Participants pretreated with MAOI iproniazid 100 mg for 4 days, 2 days off, and then IV DMT 0.35–0.83 mg/kg reported less intense psychedelic effects from DMT injection, which was a similar outcome to other studies of daily MAOI exposure reducing psychedelic effects (Sai-Halasz 1963).

The ayahuasca brew containing DMT and β-carboline harmala alkaloids is most susceptible to drug interactions since these additional harmala alkaloids are MAO-A inhibitors that significantly increase the risk of serotonin toxicity when combined with medications that also increase synaptic serotonin, such as SSRI and SNRI antidepressants, and also other medications like dextromethorphan and tramadol (Malcolm and Thomas 2021). Indeed, there has been a case report of a 36-year-old male taking fluoxetine 20 mg daily who experienced symptoms of serotonin toxicity for about 4 h after ingesting an ayahuasca brew (Callaway and Grob 1998). The harmala alkaloids are also substrates for a variety of CYP450 enzymes, such as CYP2D6, CYP1A2, CYP2C19, so enzyme inhibitor or inducers could theoretically change concentrations of harmala alkaloids (Yu et al. 2003). In addition to being MAO-A inhibitors, harmine is a CYP3A4 inhibitor, and harmine and harmaline are both CYP2D6 inhibitors, so these could also theoretically increase concentrations of concomitant drugs that are also substrates for any of these enzymes, which could subsequently increase the risk of adverse effects (Abbott et al. 2020).

5 5-MeO-DMT

5-MeO-DMT, 5-methoxy-*N,N*-dimethyltryptamine, is physiologically safe at usual doses, ~3–35 mg inhalation, ~0.7–3.1 mg/kg intravenous, and ~5–10 mg for intramuscular, sublingual and insufflation administration produce psychedelic effects (Shen et al. 2010; Uthaug et al. 2020). Injected and inhaled 5-MeO-DMT peaks within a few minutes and wears off in less than 30 min, while 5-MeO-DMT insufflated peaks around 40 min and wears off in a little over an hour (Shen et al. 2010; Uthaug et al. 2020). 5-MeO-DMT undergoes extensive metabolism through oxidative deamination by MAO-A enzymes, but also O-demethylation by CYP2D6 enzymes to active metabolite bufotenine (Ermakova et al. 2022). While MAO-A is the primary metabolic pathway for 5-MeO-DMT, there is also Phase I metabolism through N-oxygenation to 5-MeO-DMT-*N*-oxide (Ermakova et al. 2022).

5.1 5-MeO-DMT Toxicology

There has been no estimation of a usual lethal dose for humans, but estimates of LD_{50} for other species vary widely from sheep (1–5 mg/kg IV) to mice (48 mg/kg IV) (Ermakova et al. 2022). There has been a fatal case report associated with 5-MeO-DMT, but since there was co-ingestion with unknown herbal extracts and postmortem toxicology identified blood concentrations of harmine, THH, harmaline, DMT, 5-MeO-DMT, and diphenhydramine, it is difficult to determine the specific contribution of 5-MeO-DMT to the fatal toxicity (Sklerov et al. 2005; dos Santos 2013).

5.2 5-MeO-DMT Drug–Drug Interactions

5-MeO-DMT is metabolized by MAO-A and CYP2D6, so theoretically drugs that inhibit these enzymes could increase 5-MeO-DMT and active metabolite bufotenine concentrations, but this has not been evaluated in clinical trials (Ermakova et al. 2022). In animal models, when 5-MeO-DMT is co-administered with harmala alkaloids that inhibit MAO-A and CYP2D6, there have been sixfold increased exposures to 5-MeO-DMT and bufotenine (Shen et al. 2010; Jiang et al. 2013). There has also been a case report of a 17-year-old male who developed symptoms consistent with serotonin toxicity requiring emergency ICU hospitalization after he ingested an extract from three Syrian rue seeds (*Peganum harmala*; harmine +harmaline), and then smoked 5-MeO-DMT 10 mg and insufflated 5-MeO-DMT 15–20 mg powder (Brush et al. 2004). The risk of serotonin toxicity with 5-MeO-DMT may potentially be related to its very strong activation of second messenger signaling systems with 115% of the 5-HT maximum activation for 5-MeO-DMT ligand-stimulated [35S]GTPγS binding to cortical membranes, which indicated greater receptor second messenger response, including potential toxicity, than other tryptamines and 5-HT (Nonaka et al. 2007).

6 Mescaline

Mescaline, 3,4,5-thrimethoxyphenethylamine, is usually dosed between 200 and 400 mg mescaline sulfate, which is equivalent to 178–356 mg mescaline hydrochloride (Dinis-Oliveira et al. 2019). Oral mescaline plasma concentration and psychedelic effects peak after 2 h, while the duration of effects lasts about 8–12 h, with an elimination half-life of approximately 4–6 h (Dinis-Oliveira et al. 2019; Thomann et al. 2022; Ley et al. 2023). Metabolism of mescaline has not been well-characterized, but the primary pathway appears to be oxidative deamination via MAO or diamine oxidase (DAO), along with *O*-demethylation via *O*-demethylase

and *N*-acetylation, but roughly 81% is eliminated unchanged in urine within the first hour (Dinis-Oliveira et al. 2019).

6.1 Mescaline Toxicology

As dose of mescaline 8.4 g is the extrapolated usual lethal dose based on animal studies and one human case report, which is a much narrower safety margin at only 24 times greater than an average 350 mg usual effective dose consumed (Nolte and Zumwalt 1999; Gable 2004). One fatal case report was a 36-yo male with a history of alcohol use disorder who was found to have 4 separate 1 cm Mallory-Weiss esophageal lacerations with likely hemoaspiration after peyote ingestion, but the cause of death was certified as mescaline intoxication (Nolte and Zumwalt 1999).

6.2 Mescaline Drug–Drug Interactions

There have not been any well-controlled mescaline drug–drug interaction clinical trials published in the medical literature, but medications that prevent metabolism would be expected to increase the peak and duration of mescaline effects, so further research is necessary to quantify the impact of these potential drug–drug interactions.

7 2C-B

2C-B, 4-bromo-2,5dimethoxyphenethylamine, is usually dosed between 4 and 30 mg (Nugteren-van Lonkhuyzen et al. 2015). Oral fluid concentration peaks around 1 h, with an elimination half-life of 2.5 h, and duration of effects lasts about 4–8 h (Nugteren-van Lonkhuyzen et al. 2015; Papaseit et al. 2018). Metabolism of 2C-B has not been well-characterized, but the primary pathway appears to be oxidative deamination via MAO-A and MAO-B, along with demethylation that may occur via cytochrome P450 enzymes (Carmo et al. 2005).

7.1 2C-B Toxicology

There have been no fatal case reports related to 2C-B, but there have been a few case reports of severe neurological toxicity, while other analogs such as 2C-E and 2C-T-# series seem to be even more toxic with several fatal reports (Dean et al. 2013; Nugteren-van Lonkhuyzen et al. 2020). In one 2C-B case report an 18-year-old male

ingested an unknown dose and developed seizures with cerebral edema, hypertension (140/90 mmHg), tachycardia (180 bpm), hyperreflexia, and muscle rigidity indicative of serotonin toxicity with peak creatine kinase of 37,278 IU/L; the neurological impairment lasted for several months (Spoelder et al. 2019). Another seizure case report was a 19-year-old male who ingested an unknown dose of 2C-B and developed generalized tonic-clonic seizures, tachycardia (120 bpm), hypertension (175/75 mmHg), and muscle rigidity with a peak creatine kinase of 1,452 IU/L and 2C-B serum concentration at 342 ng/mL (Ho et al. 2013). The most severe case report was 43-year-old female who experienced severe headaches within 48 h of ingesting 2C-B, which was followed by upper extremity weakness several days later and eventually led to hospitalization with magnetic resonance and cerebral angiography imaging that demonstrated diffuse, toxic vasculopathy and ongoing neurologic deficits, which had not improved after 6 months follow-up (Ambrose et al. 2010).

7.2 2C-B Drug–Drug Interactions

There have not yet been any well-controlled 2C-B drug–drug interaction clinical trials published in the medical literature. In one comparative trial it appeared 2C-B duration of subjective effects was about an hour shorter than psilocybin, but any medications that inhibit MAO and relevant P450 enzymes may increase the peak and duration of 2C-B effects and further research will be necessary to quantify the impact of these potential drug–drug interactions (Mallaroni et al. 2023).

8 Bromo-DragonFLY

Bromo-DragonFLY, (1-(8-bromobenzo[1,2-b;4,5-b']difuran-4-yl)-2-aminopropane), has been dosed 200–800 μg with delay onset of effects up to 8 h and reported duration of 1–3 days (Corazza et al. 2011; Noble et al. 2018).

8.1 Bromo-DragonFLY Toxicology

Common adverse reactions may include tachycardia, hypertension, seizures, temperature fluctuations, muscle tension, arrhythmias, heart murmurs, and there have also been several international fatalities due to ischemia and end organ failure (Corazza et al. 2011; Iwersen-Bergmann et al. 2019).

8.2 Bromo-DragonFLY Drug–Drug Interactions

There have not been any well-controlled Bromo-DragonFLY drug–drug interaction clinical trials published in the medical literature, but medications that inhibit relevant substrate enzymes would be expected to increase the peak and duration of Bromo-DragonFLY effects, so further research is necessary to quantify the impact of these potential drug–drug interactions.

9 25X-NBOMe

25I-NBOMe, 2-(4-iodo-2,5-dimethoxy-phenyl)-N-[(2-methoxyphenyl)methyl] ethanamine, and other 25X-NBOMe variants have been dosed 200–1,000 μg with nasal insufflation typically requiring smaller amounts than sublingual blotter paper administration (Lawn et al. 2014). The reported duration of effects was highly variable between 3 and 13 h with a mode of 6 h for oral, sublingual, and intranasal administrations, but the peak was reported to occur earlier at 45 min for intranasal compared to about 2 h for oral or sublingual route (Lawn et al. 2014). Metabolism of 25X-NBOMe's has not been well-characterized, but the major Phase I pathway appears to mostly be via CYP3A4, but CYP2C19 and CYP2C9 also contribute to demethylation, di-demethylation, and hydroxylation, along with minor contributions from N-dealkylation; while Phase II pathway favored more rapid glucuronidation than sulfation for metabolites (Wohlfarth et al. 2017; Herian and Świt 2023).

9.1 25X-NBOMe Toxicology

In a recent literature review of 42 cases, adverse effects experienced included tachycardia (62%), hypertension (52%), diaphoresis (31%), seizures (26%), elevated creatine kinase (12%), and 6 fatal cases (14%) directly attributed to 25X-NBOMe toxicity (Lipow et al. 2022). This severe toxidrome appears to most closely resemble serotonin toxicity with profound autonomic instability in many cases (Lipow et al. 2022).

9.2 25X-NBOMe Drug–Drug Interactions

There have not been any well-controlled 25X-NBOMe drug–drug interaction clinical trials published in the medical literature.

10 Psychedelics and Serotonin Toxicity

Most classic psychedelics may cause some mild serotonin-related adverse effects, but case reports of diagnosed serotonin toxicity or death are exceptionally rare. However, novel psychoactive substances like the 25X-NBOMe compounds appear to have a more dangerous toxicological profile based on many more serotonin toxicity and fatal case reports in the medical literature. Another potential risk of serotonin toxicity would be combinations of psychedelics like 2-CB that directly increase serotonin release with ayahuasca brews that contain harmala compounds or other medications with MAOI pharmacologic effects (Malcolm and Thomas 2021). Hunter Serotonin Toxicity Criteria are the most sensitive and specific symptoms for diagnosing serotonin toxicity and include spontaneous clonus or combinations of the following: inducible clonus with agitation or diaphoresis, ocular clonus with agitation or diaphoresis, tremor and hyperreflexia, or hyperthermia (>38°C) and hypertonia with ocular or inducible clonus (Dunkley et al. 2003). Management strategies for agitation may include benzodiazepines (lorazepam 2–4 mg), while management of more severe hypertension could include IV antihypertensives like esmolol or nitroprusside (Scotton et al. 2019). The primary antidote for confirmed serotonin toxicity would be the 5-HT_{2A} antagonist cyproheptadine 12 mg with additional doses based on response, but if hyperthermia was severe (>38°C) then paralysis with vecuronium and active cooling may also be considered (Scotton et al. 2019).

11 Psychedelic Long-Term Microdosing and Valvular Heart Disease Risk

Long-term psilocybin or LSD microdosing (e.g. ~10% of a typical psychedelic dose) for many months or years may have a potential risk of valvular heart disease (VHD) due to strong 5-HT_{2B} receptor binding affinity for psilocybin's active metabolite psilocin (Ki = 4.6 nM) and LSD (Ki = 30 nM) (Halberstadt and Geyer 2011; Hutcheson et al. 2011). There have been several medications removed from the market by manufacturers due to 5-HT_{2B} receptor agonism and subsequent mitogenic effects that cause thickening of heart valve tissue and subsequent sequelae, which may include sudden cardiac death (Hutcheson et al. 2011). An FDA regulatory forum review, conducted by toxicology experts evaluating these medications associated with VHD, suggested that the best predictor of VHD was compounds with Ki values for 5-HT_{2B} close to or <10× order of magnitude relative to the Ki values for serotonin (Papoian et al. 2017). One comparable medication withdrawn from the market, pergolide, also has strong affinity for 5-HT_{2B} (Ki = 14 nM), with low dosages (<3 mg daily) and low plasma concentration (2.73 ng/mL), very similar to the psilocybin 3 mg dose with psilocin plasma concentration (2.3 ng/mL) (Blin 2003; Hutcheson et al. 2011; Madsen et al. 2019). This is concerning because MDMA (Ki = 500 nM) and its active metabolite MDA (Ki = 100 nM), both with

weaker 5-HT_{2B} affinity dosed only a few days per week, have also demonstrated a high risk of VHD in an epidemiologic study (Setola et al. 2003; Droogmans et al. 2007). In this study, 28% of MDMA users who consumed a mean of 3.6 tablets per week for 6.1 years had VHD confirmed by echocardiography versus 0% in the matched controls who had not been using MDMA (Droogmans et al. 2007). Given this theoretical risk of VHD, it will be critical to further investigate psychedelic off-target 5-HT_{2B} receptor activity in preclinical toxicology models, epidemiologic studies of psychedelic long-term microdosers, and future microdosing clinical trials.

12 Discussion and Conclusion

Most tryptamine psychedelics have a wide safety margin since the extrapolated lethal doses estimated are roughly 50 to 1,000-fold higher than the typical dose ranges. The phenethylamines, like mescaline and 2C-B, appear to have a narrower therapeutic index, while the more recent novel psychoactive substances like Bromo-DragonFLY, and especially 25X-NBOMe, may have a much greater risk of toxicity with dozens of fatal case reports (Table 1). Lethal case reports in the literature are sometimes challenging to interpret since there were usually multiple substances involved or other risk factors that make it difficult to attribute deaths to a specific psychedelic dosage. Pharmacokinetic clinical trials will be necessary to determine the clinical significance of Phase I and II metabolic enzymes related to pharmacogenetics and drug–drug interactions with these psychedelics. Most of the concomitant medications highlighted in Table 2 that desensitize or compete for the 5-HT_{2A} receptor appeared to attenuate psychedelic subjective effects, but more trials will be necessary to understand the impact of these pharmacodynamic drug–drug interactions on clinical efficacy and toxicology outcomes.

Table 1 Psychedelic toxicology

Psychedelic	Typical dose range	Extrapolated lethal dose	Lethal case analyses
LSD	50–200 μg	100 mg (freebase)	Nichols and Grob (2018)
Psilocybin	10–30 mg	6 g	Amsterdam et al. (2011) Lim et al. (2012)
DMT	25–60 mg	2 g	dos Santos (2013)
5-MeO-DMT	3–35 mg	No estimate	Sklerov (2005) dos Santos (2013)
Mescaline	200–400 mg	8.4 g	Nolte and Zumwalt (1999)
2C-B	4–30 mg	No estimate	No reports
Bromo-DragonFLY	200–800 μg	No estimate	Corazza (2011) Iwersen-Bergmann (2019)
25X-NBOMe	200–1,000 μg	No estimate	Lipow (2022)

Table 2 Psychedelic metabolism and drug–drug interactions

Psychedelic	Substrate metabolism	PK interaction changes in psychedelic exposure[a]	PD interaction changes in psychedelic effects
LSD	CYP1A2, CYP2C9, CYP2C19, CYP3A4, CYP2E1	CYP inhibitors/ inducers may alter PK profile	↓ w/ketanserin, chlorpromazine, MAOIs, SSRIs ↑ w/ergotamine, TCAs, lithium
Psilocybin	MAO-A, UGT1A9, UGT1A10	MAOIs and UGT inhibitors/inducers may alter PK profile	↓ w/ketanserin, chlorpromazine, haloperidol, risperidone, buspirone, escitalopram
DMT	MAO-A, CYP1A2, CYP2C19	MAOIs and CYP inhibitors/inducers may alter PK profile	↓ w/chronic MAOIs ↑ w/acute MAOIs, pindolol
5-MeO-DMT	MAO-A, CYP2D6	MAOIs and CYP inhibitors/inducers may alter PK profile	↑ w/acute MAOIs
Mescaline	MAO-A, MAO-B, DAO	MAOIs and DAOIs may alter PK	No studies
2C-B	MAO-A, MAO-B, and CYP enzymes	MAOIs and CYP inhibitors/inducers may alter PK profile	No studies
Bromo-DragonFLY	No studies	No studies	No studies
25X-NBOMe	CYP3A4, CYP2C9 CYP2C19	CYP inhibitors/ inducers may alter PK profile	No studies

[a] Pharmacokinetic clinical trials will be necessary to determine the clinical significance of Phase I and II metabolic enzymes related to pharmacogenetics and drug–drug interactions

References

Abbott KL, Flannery PC, Gill KS, Boothe DM, Dhanasekaran M, Mani S, Pondugula SR (2020) Adverse pharmacokinetic interactions between illicit substances and clinical drugs. Drug Metab Rev 52(1):44–65. https://doi.org/10.1080/03602532.2019.1697283

Ambrose JB, Bennett HD, Lee HS, Josephson SA (2010) Cerebral vasculopathy after 4-bromo-2,5-dimethoxyphenethylamine ingestion. Neurologist 16(3):199–202. https://doi.org/10.1097/NRL.0b013e3181a3cb53

Amsterdam J v, Opperhuizen A, Brink W v d (2011) Harm potential of magic mushroom use: a review. Regul Toxicol Pharmacol 59(3):423–429. https://doi.org/10.1016/j.yrtph.2011.01.006

Becker AM, Holze F, Grandinetti T, Klaiber A, Toedtli VE, Kolaczynska KE, Duthaler U, Varghese N, Eckert A, Grünblatt E, Liechti ME (2022a) Acute effects of psilocybin after escitalopram or placebo pretreatment in a randomized, double-blind, placebo-controlled, cross-over study in healthy subjects. Clin Pharma Ther 111(4):886–895. https://doi.org/10.1002/cpt.2487

Becker AM, Klaiber A, Holze F, Istampoulouoglou I, Duthaler U, Varghese N, Eckert A, Liechti ME (2022b) Ketanserin reverses the acute response to LSD in a randomized, double-blind, placebo-controlled, crossover study in healthy participants. Int J Neuropsychopharmacol: pyac075. https://doi.org/10.1093/ijnp/pyac075

Blin O (2003) The pharmacokinetics of pergolide in Parkinson's disease. Curr Opin Neurol 16:S9–S12. https://doi.org/10.1097/00019052-200312001-00003

Bonhaus DW, Weinhardt KK, Taylor M, Desouza A, Mcneeley PM, Szczepanski K, Fontana DJ, Trinh J, Rocha CL, Dawson MW, Flippin LA, Eglen RM (1997) RS-102221: a novel high affinity and selective, 5-HT 2C receptor antagonist. Neuropharmacology 36(4–5):621–629. https://doi.org/10.1016/S0028-3908(97)00049-X

Bonson K (1996) Chronic Administration of serotonergic antidepressants attenuates the subjective effects of LSD in humans. Neuropsychopharmacology 14(6):425–436. https://doi.org/10.1016/0893-133X(95)00145-4

Bonson KR, Murphy DL (1995) Alterations in responses to LSD in humans associated with chronic administration of tricyclic antidepressants, monoamine oxidase inhibitors or lithium. Behav Brain Res 73(1–2):229–233. https://doi.org/10.1016/0166-4328(96)00102-7

Brito-da-Costa AM, Dias-da-Silva D, Gomes NGM, Dinis-Oliveira RJ, Madureira-Carvalho Á (2020) Toxicokinetics and toxicodynamics of ayahuasca alkaloids N,N-dimethyltryptamine (DMT), harmine, harmaline and tetrahydroharmine: clinical and forensic impact. Pharmaceuticals 13(11):334. https://doi.org/10.3390/ph13110334

Brown RT, Nicholas CR, Cozzi NV, Gassman MC, Cooper KM, Muller D, Thomas CD, Hetzel SJ, Henriquez KM, Ribaudo AS, Hutson PR (2017) Pharmacokinetics of escalating doses of oral psilocybin in healthy adults. Clin Pharmacokinet 56(12):1543–1554. https://doi.org/10.1007/s40262-017-0540-6

Brush DE, Bird SB, Boyer EW (2004) Monoamine oxidase inhibitor poisoning resulting from internet misinformation on illicit substances. J Toxicol Clin Toxicol 42(2):191–195. https://doi.org/10.1081/CLT-120030949

Callaway JC, Grob CS (1998) Ayahuasca preparations and serotonin reuptake inhibitors: a potential combination for severe adverse interactions. J Psychoactive Drugs 30(4):367–369. https://doi.org/10.1080/02791072.1998.10399712

Carmo H, Hengstler JG, Boer DD, Ringel M, Remião F, Carvalho F, Fernandes E, Reys LAD, Oesch F, Bastos MDL (2005) Metabolic pathways of 4-bromo-2,5-dimethoxyphenethylamine (2C-B): analysis of phase I metabolism with hepatocytes of six species including human. Toxicology 206(1):75–89. https://doi.org/10.1016/j.tox.2004.07.004

Carter OL, Burr DC, Pettigrew JD, Wallis GM, Hasler F, Vollenweider FX (2005) Using psilocybin to investigate the relationship between attention, working memory, and the serotonin 1A and 2A receptors. https://doi.org/10.5167/UZH-121457

Carter OL, Hasler F, Pettigrew JD, Wallis GM, Liu GB, Vollenweider FX (2007) Psilocybin links binocular rivalry switch rate to attention and subjective arousal levels in humans. Psychopharmacology (Berl) 195(3):415–424. https://doi.org/10.1007/s00213-007-0930-9

Corazza O, Schifano F, Farre M, Deluca P, Davey Z, Drummond C, Torrens M, Demetrovics Z, Di Furia L, Flesland L, Mervo B, Moskalewicz J, Pisarska A, Shapiro H, Siemann H, Skutle A, Pezzolesi C, Van Der Kreeft P, Scherbaum N (2011) Designer drugs on the internet: a phenomenon out-of-control? The emergence of hallucinogenic drug bromo-dragonfly. CCP 6(2):125–129. https://doi.org/10.2174/157488411796151129

Dean BV, Stellpflug SJ, Burnett AM, Engebretsen KM (2013) 2C or not 2C: phenethylamine designer drug review. J Med Toxicol 9(2):172–178. https://doi.org/10.1007/s13181-013-0295-x

Dinis-Oliveira RJ (2017) Metabolism of psilocybin and psilocin: clinical and forensic toxicological relevance. Drug Metab Rev 49(1):84–91. https://doi.org/10.1080/03602532.2016.1278228

Dinis-Oliveira RJ, Pereira CL, Da Silva DD (2019) Pharmacokinetic and pharmacodynamic aspects of peyote and mescaline: clinical and forensic repercussions. CMP 12(3):184–194. https://doi.org/10.2174/1874467211666181010154139

dos Santos RG (2013) A critical evaluation of reports associating Ayahuasca with life-threatening adverse reactions. J Psychoactive Drugs 45(2):179–188. https://doi.org/10.1080/02791072.2013.785846

Droogmans S, Cosyns B, D'haenen H, Creeten E, Weytjens C, Franken PR, Scott B, Schoors D, Kemdem A, Close L, Vandenbossche J-L, Bechet S, Van Camp G (2007) Possible association between 3,4-methylenedioxymethamphetamine abuse and valvular heart disease. Am J Cardiol 100(9):1442–1445. https://doi.org/10.1016/j.amjcard.2007.06.045

Dunkley EJC, Isbister GK, Sibbritt D, Dawson AH, Whyte IM (2003) The hunter serotonin toxicity criteria: simple and accurate diagnostic decision rules for serotonin toxicity. QJM 96(9): 635–642. https://doi.org/10.1093/qjmed/hcg109

Ermakova AO, Dunbar F, Rucker J, Johnson MW (2022) A narrative synthesis of research with 5-MeO-DMT. J Psychopharmacol 36(3):273–294. https://doi.org/10.1177/02698811211050543

Gable RS (2004) Comparison of acute lethal toxicity of commonly abused psychoactive substances. Addiction 99(6):686–696. https://doi.org/10.1111/j.1360-0443.2004.00744.x

Halberstadt AL, Geyer MA (2011) Multiple receptors contribute to the behavioral effects of indoleamine hallucinogens. Neuropharmacology 61(3):364–381. https://doi.org/10.1016/j.neuropharm.2011.01.017

Hasler F, Bourquin D, Brenneisen R, Bär T, Vollenweider FX (1997) Determination of psilocin and 4-hydroxyindole-3-acetic acid in plasma by HPLC-ECD and pharmacokinetic profiles of oral and intravenous psilocybin in man. Pharm Acta Helv 72(3):175–184. https://doi.org/10.1016/S0031-6865(97)00014-9

Herian M, Świt P (2023) 25X-NBOMe compounds – chemistry, pharmacology and toxicology. A comprehensive review. Crit Rev Toxicol 53(1):15–33. https://doi.org/10.1080/10408444.2023.2194907

Ho RY, Gerona R, KentR O (2013) Clinical course of 4-bromo-2, 5-dimethoxyphenylamine (2C-B) intoxication with laboratory confirmation. Clin Toxicol 51(7):666–667. https://doi.org/10.3109/15563650.2013.817658

Holze F, Vizeli P, Ley L, Müller F, Dolder P, Stocker M, Duthaler U, Varghese N, Eckert A, Borgwardt S, Liechti ME (2021) Acute dose-dependent effects of lysergic acid diethylamide in a double-blind placebo-controlled study in healthy subjects. Neuropsychopharmacology 46(3): 537–544. https://doi.org/10.1038/s41386-020-00883-6

Hutcheson JD, Setola V, Roth BL, Merryman WD (2011) Serotonin receptors and heart valve disease – it was meant 2B. Pharmacol Ther 132(2):146–157. https://doi.org/10.1016/j.pharmthera.2011.03.008

Isbell H (1957) Studies on the diethylamide of lysergic acid (LSD-25): II. Effects of chlorpromazine, azacyclonol, and reserpine on the intensity of the LSD-reaction. AMA Arch NeurPsych 77(4):350. https://doi.org/10.1001/archneurpsyc.1957.02330340026002

Iwersen-Bergmann S, Lehmann S, Heinemann A, Schröder C, Müller A, Jungen H, Andresen-Streichert H, Pueschel K, Vidal C, Mercer-Chalmers-Bender K (2019) Mass poisoning with NPS: 2C-E and Bromo-DragonFly. Int J Leg Med 133(1):123–129. https://doi.org/10.1007/s00414-018-1882-9

Jiang X-L, Shen H-W, Mager DE, Yu A-M (2013) Pharmacokinetic interactions between monoamine oxidase A inhibitor harmaline and 5-methoxy-*N,N*-dimethyltryptamine, and the impact of CYP2D6 status. Drug Metab Dispos 41(5):975–986. https://doi.org/10.1124/dmd.112.050724

Keeler MH (1967) Chlorpromazine antagonism of psilocybin effect. Int J Neuropsychiatry 3:66–71

Klock JC, Boerner U, Becker CE (1975) Coma, hyperthermia, and bleeding associated with massive LSD overdose a report of eight cases. Clin Toxicol 8(2):191–203. https://doi.org/10.3109/15563657508988063

Lawn W, Barratt M, Williams M, Horne A, Winstock A (2014) The NBOMe hallucinogenic drug series: patterns of use, characteristics of users and self-reported effects in a large international sample. J Psychopharmacol 28(8):780–788. https://doi.org/10.1177/0269881114523866

Leonard JB, Anderson B, Klein-Schwartz W (2018) Does getting high hurt? Characterization of cases of LSD and psilocybin-containing mushroom exposures to national poison centers between 2000 and 2016. J Psychopharmacol 32(12):1286–1294. https://doi.org/10.1177/0269881118793086

Ley L, Holze F, Arikci D, Becker AM, Straumann I, Klaiber A, Coviello F, Dierbach S, Thomann J, Duthaler U, Luethi D, Varghese N, Eckert A, Liechti ME (2023) Comparative acute effects of mescaline, lysergic acid diethylamide, and psilocybin in a randomized, double-blind, placebo-

controlled cross-over study in healthy participants. Neuropsychopharmacology 48(11): 1659–1667. https://doi.org/10.1038/s41386-023-01607-2

Libânio Osório Marta RF (2019) Metabolism of lysergic acid diethylamide (LSD): an update. Drug Metab Rev 51(3):378–387. https://doi.org/10.1080/03602532.2019.1638931

Lim TH, Wasywich CA, Ruygrok PN (2012) A fatal case of 'magic mushroom' ingestion in a heart transplant recipient. Intern Med J 42(11):1268–1269. https://doi.org/10.1111/j.1445-5994.2012.02955.x

Lipow M, Kaleem S, Espiridion E (2022) NBOMe toxicity and fatalities: a review of the literature. Transform Med 1(1):12–18. https://doi.org/10.54299/tmed/msot8578

Luethi D, Hoener MC, Krähenbühl S, Liechti ME, Duthaler U (2019) Cytochrome P450 enzymes contribute to the metabolism of LSD to nor-LSD and 2-oxo-3-hydroxy-LSD: implications for clinical LSD use. Biochem Pharmacol 164:129–138. https://doi.org/10.1016/j.bcp.2019.04.013

Madsen MK, Fisher PM, Burmester D, Dyssegaard A, Stenbæk DS, Kristiansen S, Johansen SS, Lehel S, Linnet K, Svarer C, Erritzoe D, Ozenne B, Knudsen GM (2019) Psychedelic effects of psilocybin correlate with serotonin 2A receptor occupancy and plasma psilocin levels. Neuropsychopharmacology 44(7):1328–1334. https://doi.org/10.1038/s41386-019-0324-9

Maheux J, Éthier I, Rouillard C, Lévesque D (2005) Induction patterns of transcription factors of the *Nur* Family (*Nurr1*, *Nur77*, and *Nor*-1) by typical and atypical antipsychotics in the mouse brain: implication for their mechanism of action. J Pharmacol Exp Ther 313(1):460–473. https://doi.org/10.1124/jpet.104.080184

Malcolm B, Thomas K (2021) Serotonin toxicity of serotonergic psychedelics. Psychopharmacology (Berl). https://doi.org/10.1007/s00213-021-05876-x

Mallaroni P, Mason NL, Reckweg JT, Paci R, Ritscher S, Toennes SW, Theunissen EL, Kuypers KPC, Ramaekers JG (2023) Assessment of the acute effects of 2C-B vs psilocybin on subjective experience, mood, and cognition. Clin Pharma Therapeutics 114(2):423–433. https://doi.org/10.1002/cpt.2958

Manevski N, Kurkela M, Höglund C, Mauriala T, Court MH, Yli-Kauhaluoma J, Finel M (2010) Glucuronidation of psilocin and 4-hydroxyindole by the human UDP-glucuronosyltransferases. Drug Metab Dispos 38(3):386–395. https://doi.org/10.1124/dmd.109.031138

Mattusek N, Halbach A (1964) Über den Einfluss von Ergotamin auf die LSD-"Psychose" beim menschen. [on the influence of ergotamine on LSD-"Psychosis" in man]. In: Psychopharmacologia. pp 158–160

Murphree HB (1962) Quantitative studies in humans on the antagonism of lysergic acid diethylamide by chlorpromazine and phenoxybenzamine. Clin Pharmacol Ther 3(3):314–320. https://doi.org/10.1002/cpt196233314

Nayak S, Gukasyan N, Barrett FS, Erowid E, Erowid F, Griffiths RR (2021) Classic psychedelic coadministration with lithium, but not lamotrigine, is associated with seizures: an analysis of online psychedelic experience reports

Nichols DE (2016) Psychedelics. Pharmacol Rev 68(2):264–355. https://doi.org/10.1124/pr.115.011478

Nichols DE (2020) Psilocybin: from ancient magic to modern medicine. J Antibiot 73(10):679–686. https://doi.org/10.1038/s41429-020-0311-8

Nichols DE, Grob CS (2018) Is LSD toxic? Forensic Sci Int 284:141–145. https://doi.org/10.1016/j.forsciint.2018.01.006

Noble C, Holm NB, Mardal M, Linnet K (2018) Bromo-dragonfly, a psychoactive benzodifuran, is resistant to hepatic metabolism and potently inhibits monoamine oxidase a. Toxicol Lett 295: 397–407. https://doi.org/10.1016/j.toxlet.2018.07.018

Nolte KB, Zumwalt RE (1999) Fatal peyote ingestion associated with Mallory-Weiss lacerations. West J Med 170(6):328

Nonaka R, Nagai F, Ogata A, Satoh K (2007) In vitro screening of psychoactive drugs by [35S] GTP.GAMMA.S binding in rat brain membranes. Biol Pharm Bull 30(12):2328–2333. https://doi.org/10.1248/bpb.30.2328

Nugteren-van Lonkhuyzen JJ, Van Riel AJHP, Brunt TM, Hondebrink L (2015) Pharmacokinetics, pharmacodynamics and toxicology of new psychoactive substances (NPS): 2C-B, 4-fluoroamphetamine and benzofurans. Drug Alcohol Depend 157:18–27. https://doi.org/10.1016/j.drugalcdep.2015.10.011

Nugteren-van Lonkhuyzen JJ, De Lange DW, Van Riel AJHP, Vrolijk RQ, Ohana D, Hondebrink L (2020) The clinical toxicology of 4-bromo-2,5-dimethoxyphenethylamine (2C-B): the severity of poisoning after exposure to low to moderate and high doses. Ann Emerg Med 76(3):303–317. https://doi.org/10.1016/j.annemergmed.2020.04.022

Olbrich S, Preller KH, Vollenweider FX (2021) LSD and ketanserin and their impact on the human autonomic nervous system. Psychophysiology 58(6). https://doi.org/10.1111/psyp.13822

Papaseit E, Farré M, Pérez-Mañá C, Torrens M, Ventura M, Pujadas M, De La Torre R, González D (2018) Acute pharmacological effects of 2C-B in humans: an observational study. Front Pharmacol 9:206. https://doi.org/10.3389/fphar.2018.00206

Papoian T, Jagadeesh G, Saulnier M, Simpson N, Ravindran A, Yang B, Laniyonu AA, Khan I, Szarfman A (2017) Regulatory forum review*: utility of *in vitro* secondary pharmacology data to assess risk of drug-induced valvular heart disease in humans: regulatory considerations. Toxicol Pathol 45(3):381–388. https://doi.org/10.1177/0192623317690609

Pokorny T, Preller KH, Kraehenmann R, Vollenweider FX (2016) Modulatory effect of the 5-HT1A agonist buspirone and the mixed non-hallucinogenic 5-HT1A/2A agonist ergotamine on psilocybin-induced psychedelic experience. Eur Neuropsychopharmacol 26(4):756–766. https://doi.org/10.1016/j.euroneuro.2016.01.005

Resnick O, Krus DM, Raskin M (1964) LSD-25 action in normal subjects treated with a monoamine oxidase inhibitor. Life Sci 3(11):1207–1214. https://doi.org/10.1016/0024-3205(64)90037-2

Riba J, McIlhenny EH, Valle M, Bouso JC, Barker SA (2012) Metabolism and disposition of N, N-dimethyltryptamine and harmala alkaloids after oral administration of ayahuasca: metabolism and disposition of ayahuasca alkaloids. Drug Test Anal 4(7–8):610–616. https://doi.org/10.1002/dta.1344

Sai-Halasz A (1963) The effect of MAO inhibition on the experimental psychosis induced by dimethyltryptamin. Psychopharmacologia 4(5):385–388. https://doi.org/10.1007/BF00405249

Sarparast A, Thomas K, Malcolm B, Stauffer CS (2022) Drug-drug interactions between psychiatric medications and MDMA or psilocybin: a systematic review. Psychopharmacology (Berl) 239(6):1945–1976. https://doi.org/10.1007/s00213-022-06083-y

Scotton WJ, Hill LJ, Williams AC, Barnes NM (2019) Serotonin syndrome: pathophysiology, clinical features, management, and potential future directions. Int J Tryptophan Res 12: 117864691987392. https://doi.org/10.1177/1178646919873925

Setola V, Hufeisen SJ, Grande-Allen KJ, Vesely I, Glennon RA, Blough B, Rothman RB, Roth BL (2003) 3,4-Methylenedioxymethamphetamine (MDMA, "Ecstasy") induces Fenfluramine-like proliferative actions on human cardiac valvular interstitial cells in vitro. Mol Pharmacol 63(6): 1223–1229. https://doi.org/10.1124/mol.63.6.1223

Shen H-W, Jiang X-L, Winter JC, Yu A-M (2010) Psychedelic 5-methoxy-N,N-dimethyltryptamine: metabolism, pharmacokinetics, drug interactions, and pharmacological actions. CDM 11(8):659–666. https://doi.org/10.2174/138920010794233495

Sklerov J, Levine B, Moore KA, King T, Fowler D (2005) A Fatal intoxication following the ingestion of 5-methoxy-N,N-dimethyltryptamine in an Ayahuasca Preparation*. J Anal Toxicol 29(8):838–841. https://doi.org/10.1093/jat/29.8.838

Spoelder AS, Louwerens JKG, Krens SD, Jager N, LeCouffe NE, Ruijter W, Brunt TM (2019) Unexpected serotonin syndrome, epileptic seizures, and cerebral edema following 2,5-dimethoxy-4-bromophenethylamine ingestion. J Forensic Sci 64(6):1950–1952. https://doi.org/10.1111/1556-4029.14214

Strassman RJ (1995) Human psychopharmacology of N,N-dimethyltryptamine. Behav Brain Res 73(1–2):121–124. https://doi.org/10.1016/0166-4328(96)00081-2

Thomann J, Ley L, Klaiber A, Liechti ME, Duthaler U (2022) Development and validation of an LC-MS/MS method for the quantification of mescaline and major metabolites in human plasma. J Pharm Biomed Anal 220:114980. https://doi.org/10.1016/j.jpba.2022.114980

Uthaug MV, Lancelotta R, Ortiz Bernal AM, Davis AK, Ramaekers JG (2020) A comparison of reactivation experiences following vaporization and intramuscular injection (IM) of synthetic 5-methoxy-N,N-dimethyltryptamine (5-MeO-DMT) in a naturalistic setting. J Psychedelic Stud 4(2):104–113. https://doi.org/10.1556/2054.2020.00123

Van Oekelen D, Luyten WHML, Leysen JE (2003) 5-HT2A and 5-HT2C receptors and their atypical regulation properties. Life Sci 72(22):2429–2449. https://doi.org/10.1016/S0024-3205(03)00141-3

Vizeli P, Straumann I, Holze F, Schmid Y, Dolder PC, Liechti ME (2021) Genetic influence of CYP2D6 on pharmacokinetics and acute subjective effects of LSD in a pooled analysis. Sci Rep 11(1):10851. https://doi.org/10.1038/s41598-021-90343-y

Vollenweider FX, Vollenweider-Scherpenhuyzen MFI, Bäbler A, Vogel H, Hell D (1998) Psilocybin induces schizophrenia-like psychosis in humans via a serotonin-2 agonist action. Neuroreport 9(17):3897–3902. https://doi.org/10.1097/00001756-199812010-00024

Wagmann L, Meyer MR, Maurer HH (2016) What is the contribution of human FMO3 in the N-oxygenation of selected therapeutic drugs and drugs of abuse? Toxicol Lett 258:55–70. https://doi.org/10.1016/j.toxlet.2016.06.013

Wagmann L, Richter LHJ, Kehl T, Wack F, Bergstrand MP, Brandt SD, Stratford A, Maurer HH, Meyer MR (2019) In vitro metabolic fate of nine LSD-based new psychoactive substances and their analytical detectability in different urinary screening procedures. Anal Bioanal Chem 411(19):4751–4763. https://doi.org/10.1007/s00216-018-1558-9

Wohlfarth A, Roman M, Andersson M, Kugelberg FC, Diao X, Carlier J, Eriksson C, Wu X, Konradsson P, Josefsson M, Huestis MA, Kronstrand R (2017) 25C-NBOMe and 25I-NBOMe metabolite studies in human hepatocytes, *in vivo* mouse and human urine with high-resolution mass spectrometry: metabolism of 25C- and 25I-NBOMe. Drug Test Anal 9(5):680–698. https://doi.org/10.1002/dta.2044

Yamauchi M, Miyara T, Matsushima T, Imanishi T (2006) Desensitization of 5-HT2A receptor function by chronic administration of selective serotonin reuptake inhibitors. Brain Res 1067(1):164–169. https://doi.org/10.1016/j.brainres.2005.10.075

Yu A-M, Idle JR, Krausz KW, Küpfer A, Gonzalez FJ (2003) Contribution of individual cytochrome P450 isozymes to the *O*-demethylation of the psychotropic β-carboline alkaloids harmaline and harmine. J Pharmacol Exp Ther 305(1):315–322. https://doi.org/10.1124/jpet.102.047050